DATA SCIENCE REPORT

MANUFACTURING DEFECT INTELLIGENCE: DEFECT SEVERITY CLASSIFICATION & QUALITY ANALYTICS

Defect Severity, Quality Analytics & Predictive Modeling



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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Overview & Business Objective

1.1 PROJECT OVERVIEW

Manufacturing quality assurance plays a critical role in maintaining product reliability, identifying defects, and supporting effective inspection and repair decisions. Defects vary in type, location, and severity, making accurate severity classification an important aspect of quality management.

Understanding these defect characteristics can help manufacturers assess quality risks, prioritize inspections, and improve defect management processes.

The Manufacturing Defect Intelligence project applies machine learning to manufacturing defect records to analyze defect patterns and classify severity into Critical, Minor, and Moderate categories.

The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Behavioral feature engineering
- Class imbalance handling
- Machine learning classification
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to investigate the feasibility of predicting defect severity, but also to evaluate classification reliability, understand the factors influencing model predictions, and identify opportunities for improving manufacturing quality assurance.

1.2 PROBLEM STATEMENT

Manufacturing organizations must evaluate defects and determine their severity to support inspection prioritization, repair planning, and quality management.

Severity assessment may depend on multiple factors, including defect characteristics, defect location, inspection procedures, and manufacturing conditions. When these relationships are complex or insufficiently documented severity classification can become challenging.

The central problem addressed in this project is:

Can manufacturing defect characteristics and inspection information be used to reliably classify defects into Critical, Minor, and Moderate severity levels, while identifying the limitations and factors influencing model predictions?

The project formulates this task as a supervised multiclass classification problem, with ***severity*** as the target variable.

Particular attention is given to the identification of Critical defects, where incorrect predictions may have greater operational consequences than ordinary classification errors.

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Examine the structure, quality, and characteristics of the manufacturing defects dataset.
- Investigate defect distributions, severity patterns, inspection methods, and repair costs through exploratory data analysis.
- Identify suitable predictive variables while excluding identifiers and potentially leakage-prone features.
- Engineer temporal features from recorded defect dates.
- Develop a reproducible preprocessing workflow for numerical, categorical, and binary variables.
- Train and compare multiple machine learning classification algorithms.
- Optimize selected models through hyperparameter tuning and stratified cross-validation.
- Investigate whether voting and stacking ensembles improve classification performance.
- Evaluate model generalization using a separate held-out test dataset.
- Examine Critical defect recall, false alerts, and missed severe defects.
- Explain model behavior using coefficients, feature importance, permutation analysis, and SHAP.
- Translate the findings into practical quality assurance considerations and identify opportunities for future improvement.

1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, behavioral feature engineering, machine learning, model evaluation, and explainability to understand purchase patterns, identify key conversion signals, evaluate predictive performance, and derive actionable e-commerce insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Defect distribution analysis
- Severity and repair cost analysis

02

Predictive Modeling

- Feature preparation
- Temporal feature engineering
- Multiclass classification
- Hyperparameter tuning and ensemble learning

03

Explainability

- Model performance comparison
- Critical defect detection analysis
- Error analysis & SHAP Analysis

04

Business Interpretation

- Defect severity insights
- Inspection and quality risk assessment
- Classification reliability analysis
- Recommendations for quality improvement

Data Understanding & Exploratory Data Analysis

2.1 DATA OVERVIEW

The project uses the Manufacturing Defects Dataset, a synthetic dataset containing records of manufacturing defects and their associated characteristics.

Data Source: Kaggle

Dataset Size

1,000
Records

8
Original Features

Purchase Diversity

3
Severity Levels

3
Inspection Methods

3
Defect Types



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
defect_id	Categorical (Identifier)	Unique identifier assigned to a defect record
product_id	Categorical (Identifier)	Identifier of the associated product
defect_type	Categorical (Nominal)	Classification of the manufacturing defect
defect_date	Date	Date associated with the recorded defect
defect_location	Categorical (Nominal)	Location of the defect within the product
severity	Categorical (Ordinal)	Severity classification of the defect (Target Variable)
inspection_method	Categorical (Nominal)	Method used to inspect or identify the defect
repair_cost	Numerical (Continuous)	Recorded cost associated with defect repair

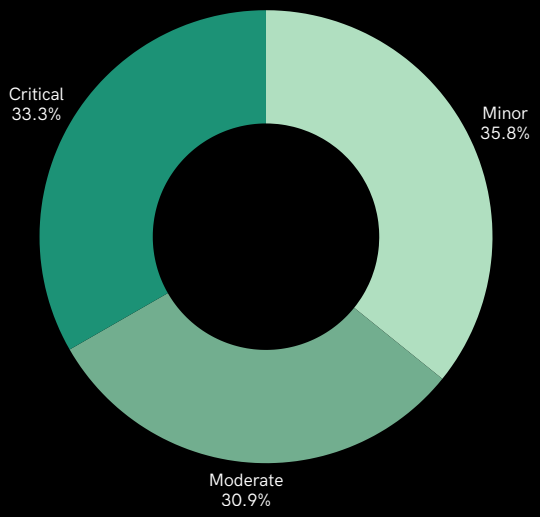
2.2 TARGET VARIABLE

The target variable selected for classification is:

severity

It represents the recorded severity classification of each manufacturing defect.

The dataset contains three severity categories:



The severity classes are relatively balanced, with only modest differences in representation.

This is useful for multiclass modeling because no single class overwhelmingly dominates the dataset. Nevertheless, balanced class frequencies do not guarantee that the classes are distinguishable using the available features.

2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Target variable distribution
- Potential outliers
- Feature relationships
- Class imbalance

No missing values were identified in the dataset.

Several variables required particular consideration before model development.

defect_id was excluded because it serves as a unique record identifier rather than a meaningful description of defect severity.

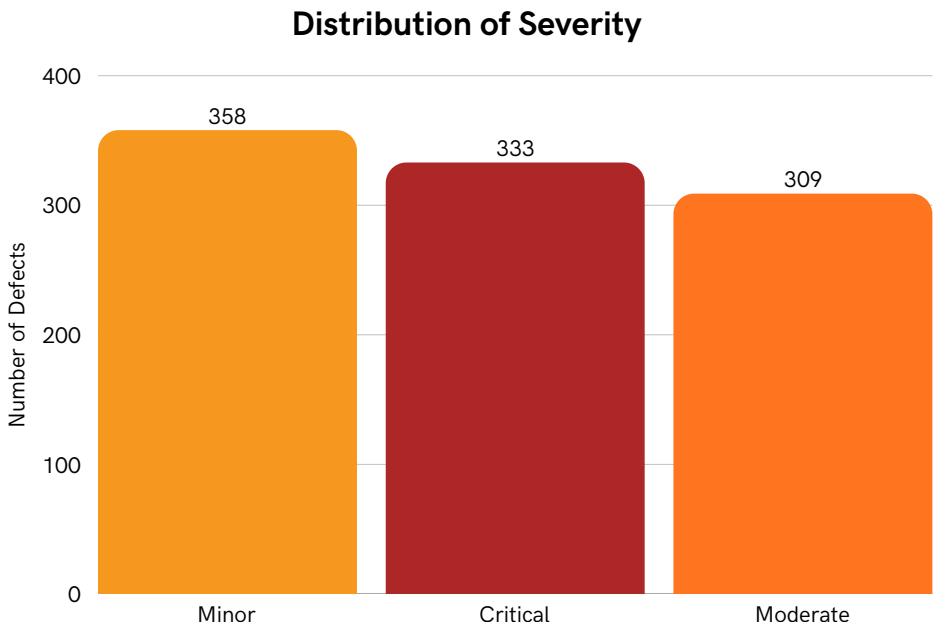
product_id was excluded because of its high cardinality and the absence of additional product-level information supporting its use.

repair_cost was excluded from predictive modeling because repair costs may be determined after defect severity assessment. Including such information could create target leakage in a system intended to classify defects before repair decisions are made.

2.4 Exploratory Data Analysis

Defect severity is relatively evenly distributed, with Minor defects accounting for the largest share

~25%
New Visitor Conversion



Key observations

- Minor defects account for 35.8% of observations, followed by Critical defects at 33.3% and Moderate defects at 30.9%.
- The three severity classes have relatively similar frequencies, indicating no substantial class imbalance.

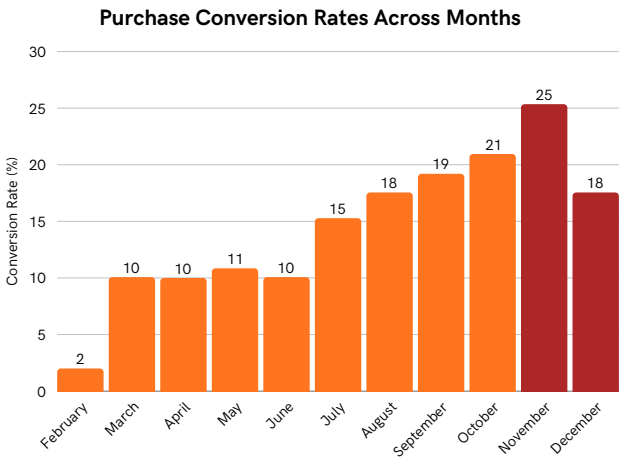
Business Insights

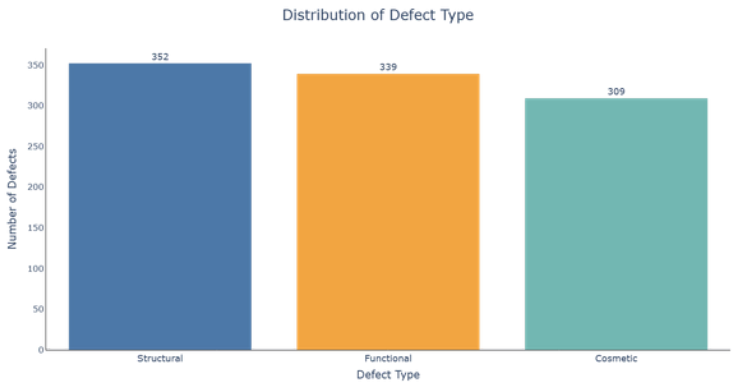
- Critical defects represent approximately one-third of recorded defects, making their identification an important quality assurance priority.
- The balanced distribution supports comparisons across severity categories, although accurate classification still depends on the availability of informative predictors.

Distribution of Defect Type

25.35%

November Conversion Rate





Key observations

- Structural defects account for 352 observations (35.2%), followed by Functional defects at 339 (33.9%) and Cosmetic defects at 309 (30.9%).
- All three defect categories have comparable representation, with Structural defects occurring slightly more frequently.

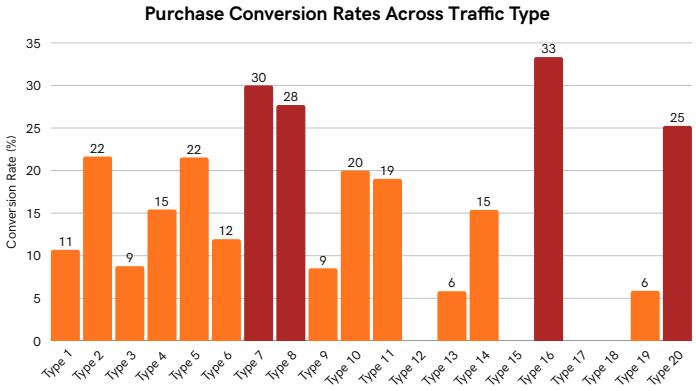
Business Insights

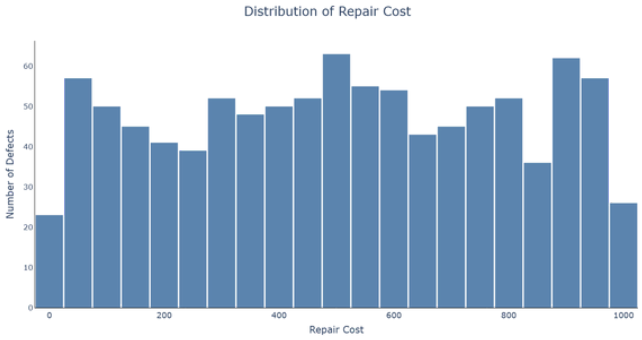
- Quality monitoring should consider structural, functional, and cosmetic defects rather than concentrating exclusively on one category.
- The relatively even distribution provides a basis for comparing defect characteristics and severity patterns across defect types.

Distribution of Repair Cost

33.33%

Traffic Type 16 Conversion





Key observations

- Repair costs extend across a broad range, approximately from 10 to 1,000.
- The distribution appears broadly uniform, without a pronounced concentration in a particular cost interval.

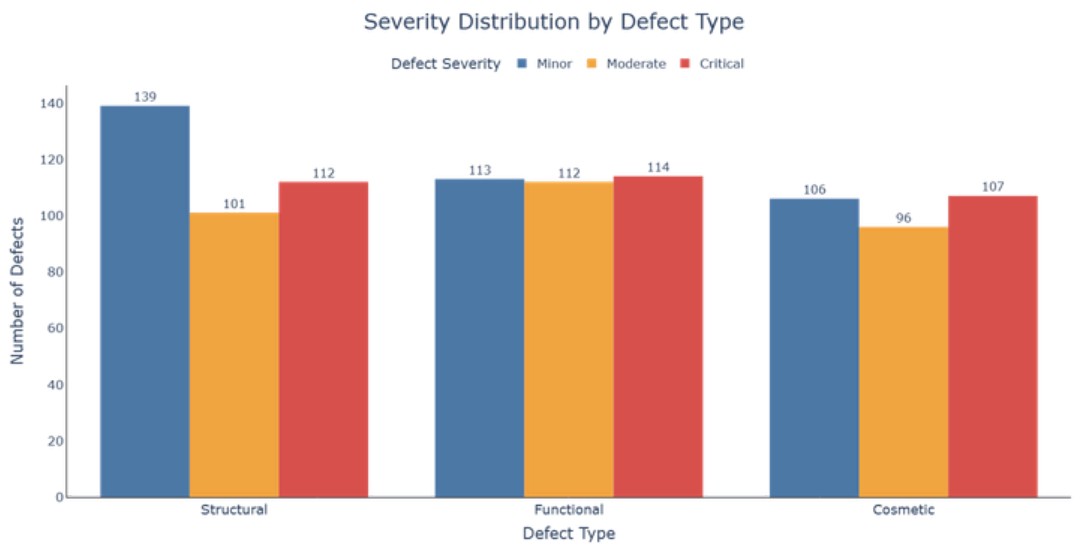
Business Insights

- The considerable variation in repair costs highlights the importance of understanding repair expenditure when planning manufacturing quality management activities.
- Additional operational information may be required to explain why some defects incur substantially higher repair costs than others.

Severity Distribution by Defect Type

~35

Product Pages Viewed (median)



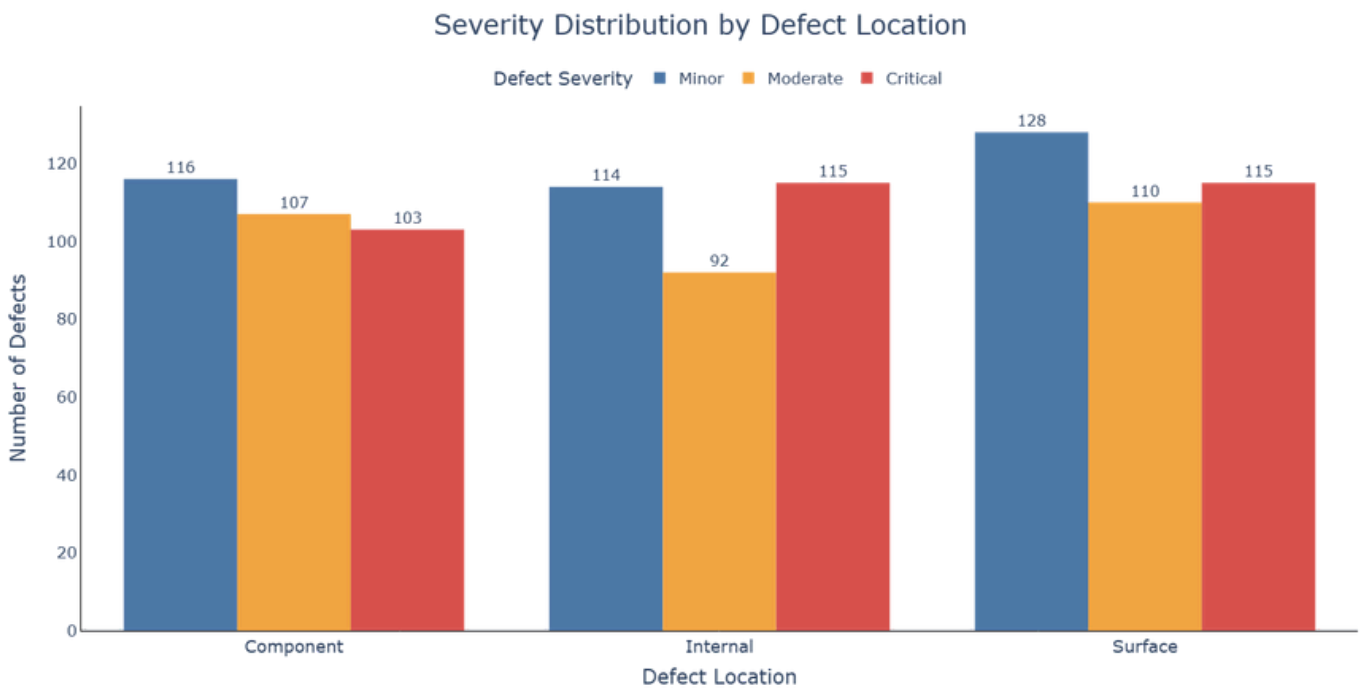
Key observations

- Structural defects have the highest Minor defect count, while Functional defects show nearly equal representation across all three severity categories.
- Critical defects occur across Structural, Functional, and Cosmetic categories without a pronounced concentration in one type.

Business Insights

- Defect type alone may provide limited information for distinguishing severity levels.
- Quality assurance procedures should evaluate severity within every defect category rather than assuming a particular defect type is inherently more critical.

Severity Distribution by Defect Location



Key observations

- Surface defects have the highest Minor defect count, while Internal defects show nearly equal Minor and Critical counts.
- Critical defects occur across Component, Internal, and Surface locations at relatively similar frequencies.

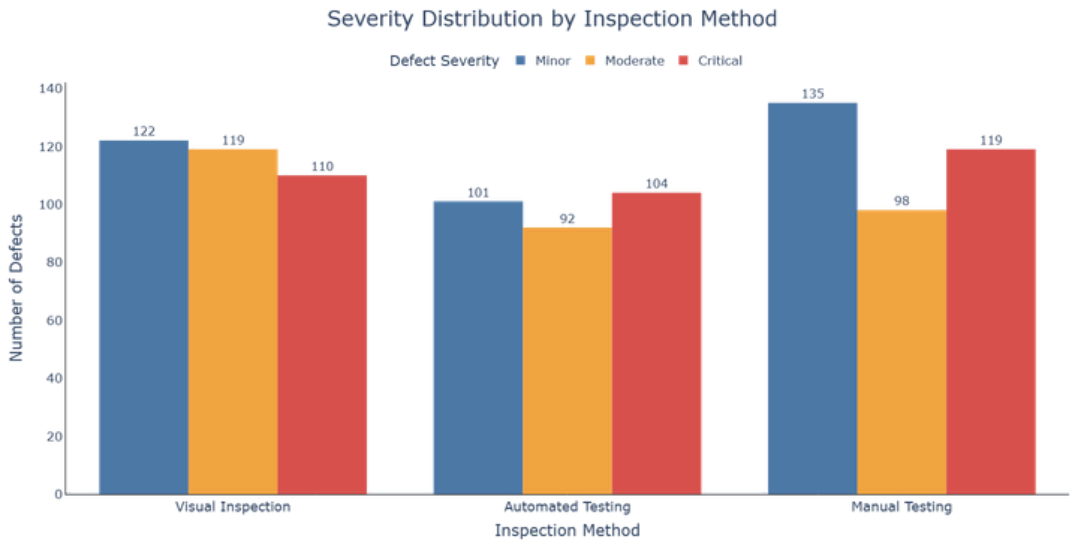
Business Insights

- Inspection coverage should include all defect locations because Critical defects are not restricted to a single area.
- Defect location may provide useful descriptive information but does not show strong visual separation between severity classes.

Severity Distribution by Inspection Method

52.03%

Typical CPU Utilization (Median)



Key observations

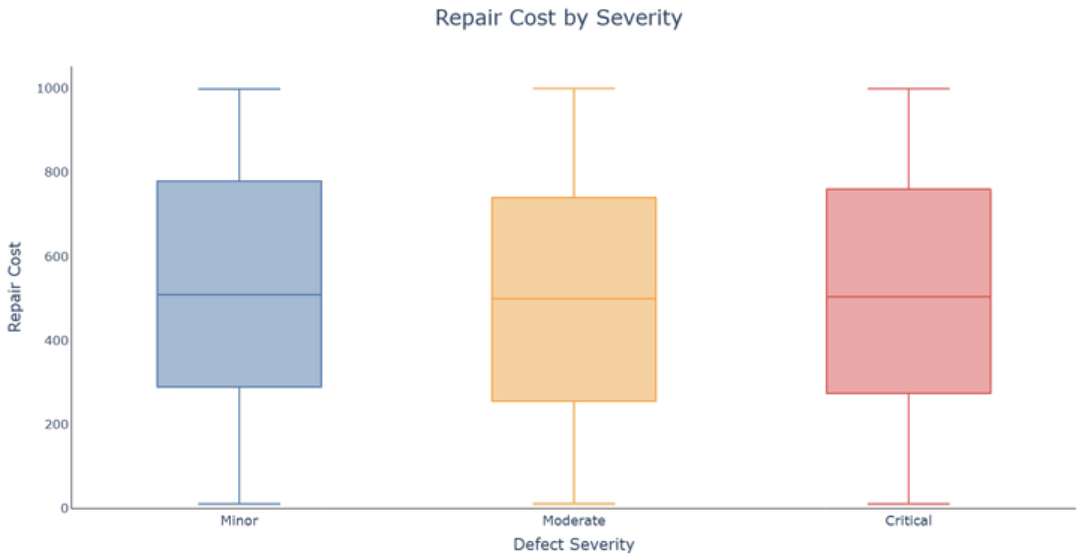
- Manual Testing records the highest Minor defect count, while Visual Inspection shows relatively similar counts across severity categories.
- Automated Testing has slightly more Critical than Minor defects, although differences between severity counts remain modest.

Business Insights

- All three inspection methods are associated with the identification of Critical, Minor, and Moderate defects.
- Differences in recorded counts should not be interpreted as evidence of inspection effectiveness without accounting for inspection volumes and detection conditions.

Repair Cost by Severity

0.0029
Median Bounce Rate



Key observations

- Median repair costs are approximately 500 across Minor, Critical, and Moderate defects.
- All three severity categories exhibit substantial cost variation and strongly overlapping distributions.

Business Insights

- Repair cost does not show a clear increasing pattern with defect severity in this dataset.
- Severity-based repair budgeting may require additional operational information beyond the recorded severity classification.

PageValues has a strong positive relationship with purchase conversion, while BounceRates ExitRates show negative relationships.



Key observations

- Repair costs vary widely within every combination of defect type and severity, with substantial overlap between groups.
- Median costs differ somewhat across combinations, but no consistent pattern indicates that Critical defects always incur higher repair costs.

Business Insights

- Neither defect type nor severity appears sufficient on its own to explain repair cost differences.
- More detailed information about repair procedures, defect complexity, and resource requirements would help investigate cost drivers.